

MAKE UP EXAMINATIONS – APRIL 2025

Program	: B.E. – CSE (Cyber Security)	Semester	: III
Course Name	: Computer Organization and Architecture	Max. Marks	: 100
Course Code	: CY34	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Simplify the following four variable expression $Y = \sum m(0,2,6,7,8,15)$
*Note: Student can use K-Map or Boolean Algebra approach. CO1 (08)
 - List the difference between combinational circuits and Sequential circuits. With the help of Truth -Table design 4: 1 Mux circuit. CO1 (06)
 - Explain clocked JK flip-flop with neat circuit diagram. CO1 (06)
- Simplify the following three variable expression using Boolean algebra. $Y = \sum m(1,2,3)$ and $Y = \prod M(0,2,6,7)$ CO1 (08)
 - Implement Boolean expression for Ex-OR gate using NOR gates. CO1 (06)
 - List the basic types of shift registers in terms of the data movement and explain any two. CO1 (06)

UNIT - II

- With a neat diagram explain the indirect addressing mode. Write a sequence of machine instruction to add N elements stored in the memory using the same. CO2 (08)
 - Explain the basic performance equation and its parameters. Explain the different ways of optimizing these parameters to improve the performance of a computer. CO2 (06)
 - What are the steps involved in sequencing instructions during the fetch-decode-execute cycle? CO2 (06)
- Demonstrate how the operation $P = (Q * R - S / T) + U * V$ can be implemented in a
 - One address
 - Two address and
 - Three address instructions
CO2 (08)
 - What is subroutine linkage method? Explain the execution effect of call and return instructions with respect to the subroutine nesting. CO2 (06)
 - Illustrate the working of single bus organization of a processor with a neat diagram. CO2 (06)

UNIT - III

- With the help of a neat diagram, describe how multiple devices are connected in a daisy chain configuration to handle interrupts. Explain how the priority of devices is determined and how an interrupt request is passed along the chain. CO3 (08)

- b) Compare synchronous buses and asynchronous buses in a computer system in terms of speed, complexity, scalability, and cost. CO3 (06)
- c) In a binary representation, a 4-bit number value is 1111. What is the value it represents in (I) sign-magnitude, (II) 1's complement, and (III) 2's complement. CO3 (06)
- 6. a) With neat diagram(s), describe the steps involved in a DMA data transfer between an I/O device and memory. CO3 (08)
- b) Compare and contrast DMA-controlled and interrupt-driven methods for transferring data from an input device to memory. CO3 (06)
- c) Design a 4-bit carry lookahead adder. CO3 (06)

UNIT- IV

- 7. a) What are the main differences between Semiconductor RAM (SRAM and DRAM) and ROM (Read-Only Memory)? Discuss their respective uses in a computer system. CO4 (08)
- b) Explain the concept of cache mapping functions. Describe the Direct Mapping technique in cache memory with an example. CO4 (06)
- c) Explain the basic concept of cache memory and its role in improving system performance. CO4 (06)
- 8. a) Discuss cache replacement algorithms (FIFO, LRU, and Optimal). Explain how the choice of a replacement algorithm affects the performance of the cache memory system. CO4 (07)
- b) Explain the trade-offs between speed, size, and cost in the design of memory systems. CO4 (07)
- c) Explain the process of multiplying two positive integers using the standard multiplication algorithm. CO4 (06)

UNIT – V

- 9. a) List out the key aspects of GPU programs driving GPU processor architecture and explain any one in briefly. CO5 (10)
- b) Discuss the performance of parallel computers based on Amdahl's Law. CO5 (10)
- 10. a) List out the difference of CISC and RISC style Processors. CO5 (08)
- b) Discuss the operation of four stage pipeline. CO5 (06)
- c) Illustrate along with key characteristic as how GPUs vary from CPUs. CO5 (06)
